Automated PCB Defect Detection Using Computer Vision

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Computer Vision
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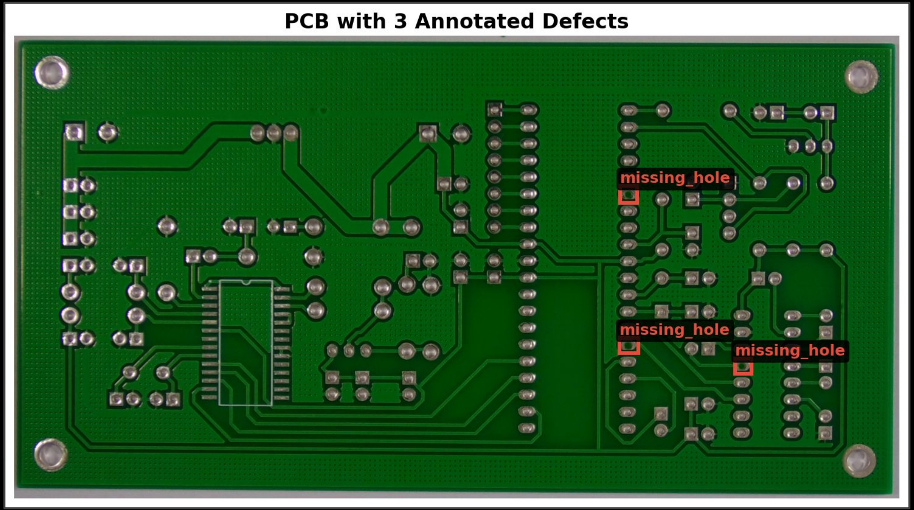
Executive Summary

- AI-powered inspection of PCB manufacturing defects
- 3 model approaches compared across 1,386 images and 6 defect types
- Missed defects cost 10x more than false alarms
- Real-time detection enables scalable, automated quality assurance



PCB with 3 Annotated Defects

The Team

Matt Miller

Software engineer with 15+ years experience.

Pursuing Master's in Data science from Northwestern University

The Problem

- Manual PCB inspection is slow, subjective and doesn't scale
- 6 defect types: missing holes, mouse bites, open circuits, shorts, spurs, spurious copper
- A missed defect costs ~$500 (warranty) vs. ~$50 for a false alarm*
- Industry needs fast, consistent, automated visual inspection

** Source: 1-10-100 Rule (Labovitz & Chang, 1993); cost escalation validated by NASA/INCOSE (Stecklein et al., 2004)*

The Cost of Getting It Wrong

~$50/board

False Alarm (unnecessary rework)

~$500/board

Missed defect (warranty claim)

Priority

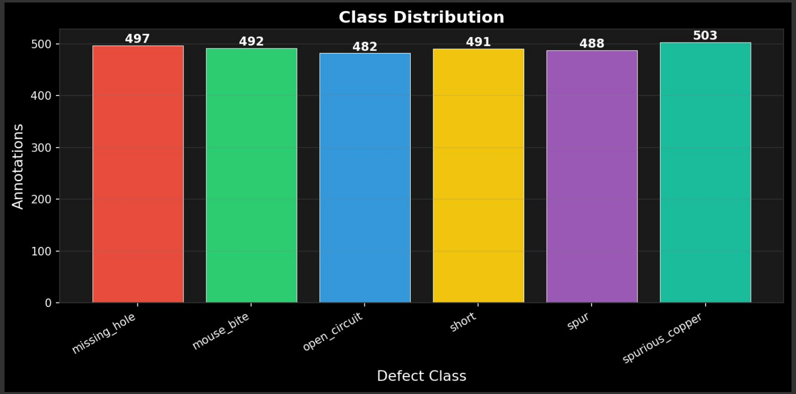
System must prioritize **recall**; capturing all defects.

"[F]ixing a defect during production is far cheaper than addressing it after assembly or in the field, with rework costs reducible by up to 80%"*

* Source: AllPCB, "The Hidden Costs of Poor PCB Quality: A Financial Impact Analysis"

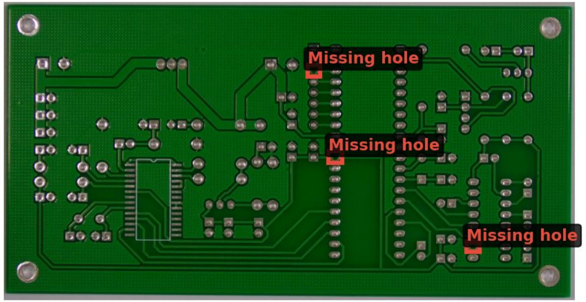
- **1,386 PCB images** with bounding box annotations (Kaggle)
- **1,500 paired template/defective images** (DeePCB)
- 6 balanced defect classes (~231 images each)
- Two annotation formats (Pascal VOC XML + custom TXT) unified into a single standard
- Split: 70% train / 15% validation / 15% test

Defect Class Distribution

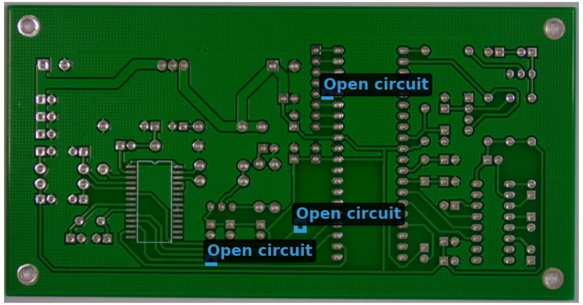


The Six Defect Types

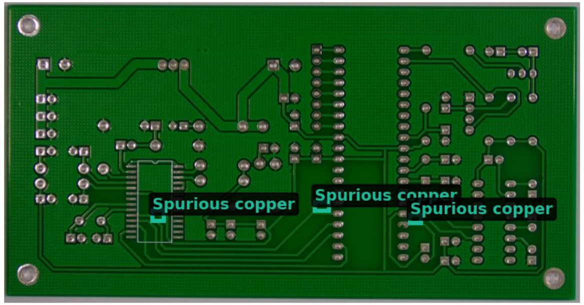
Missing hole



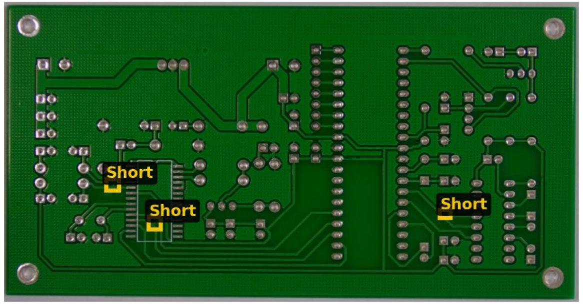
Open circuit



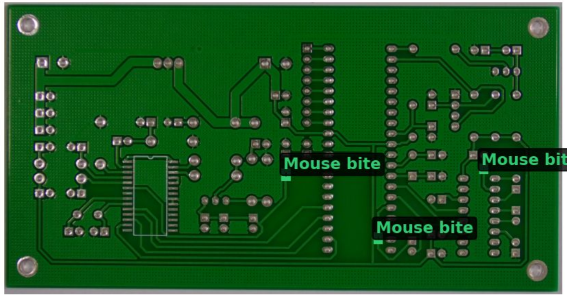
Spurious copper



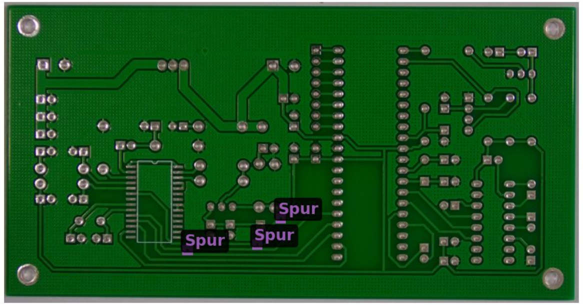
Short



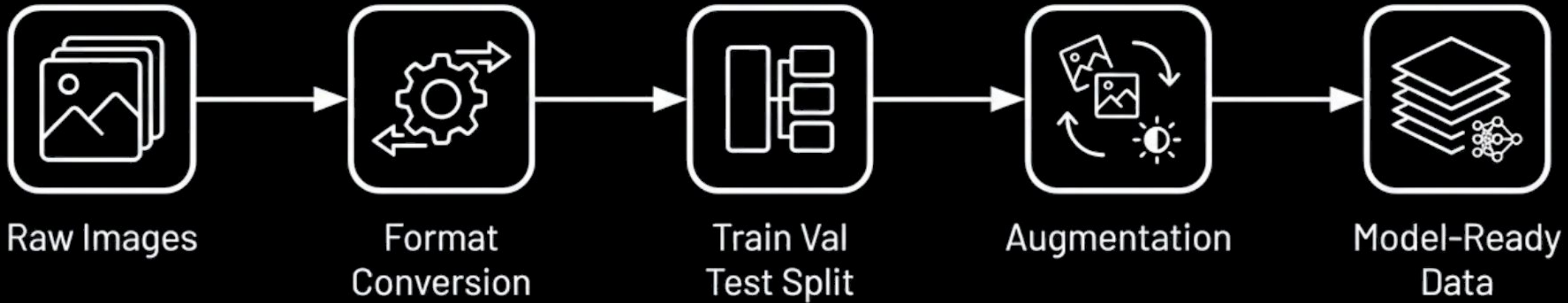
Mouse bite



Spur



Preparing the Data



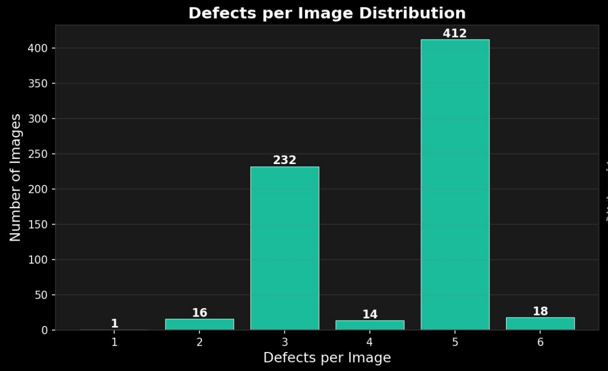
Exploratory Data Analysis

1
Balanced Classes
Near-uniform distribution (~490 annotations each). No oversampling needed.

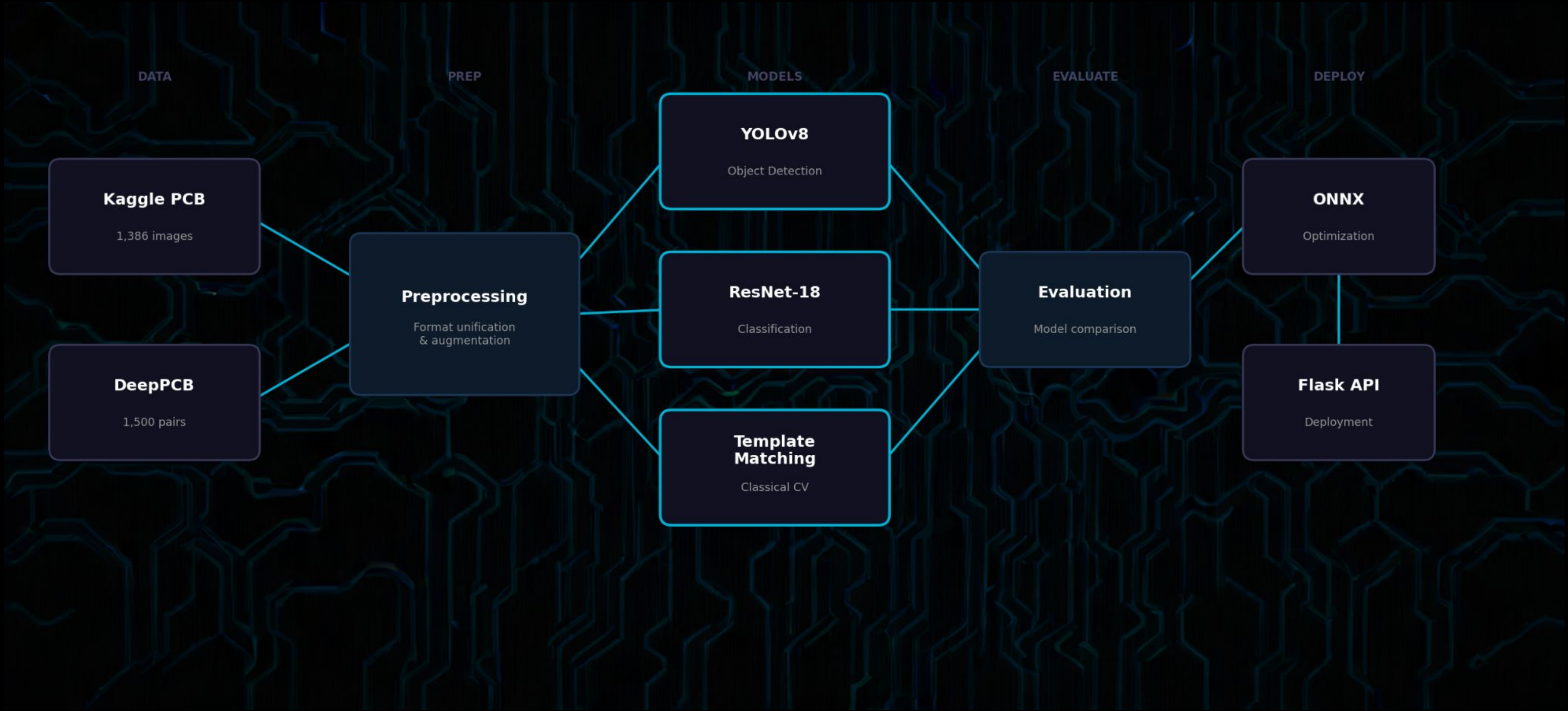
2
High Resolution
Size avg 2778×2138px, resized to 640×640 for YOLOv8. Small defects risk being lost.

3
Multi-Defect
Most images contain 4-5 defects. Model must handle overlapping / adjacent bounding boxes.

4
Single Board Design
All images from one PCB layout with synthetic defects. Model needs fine-tuning for new designs.



Solution Architecture



Technology Stack

Detection

YOLOv8 (Ultralytics)

PyTorch

COCO Transfer Learning

Classification

ResNet-18

PyTorch

ImageNet transfer learning

Classical CV

OpenCV

ORB feature matching

Template differencing

Optimization

ONNX Runtime

INT8 quantization

Deployment

Flask REST API

JSON inference endpoint

Infrastructure

Apple M4 (MPS GPU)

GitHub Pages

Jupyter Book

Methodology

YOLOv8

- Detects and classifies defects in a single pass (~35ms/image)
- Pretrained on 80-class COCO dataset, fine-tuned on our 6 PCB defect types

ResNet-18

- Validates YOLO's defect type labels.
- 98.1% agreement - confirms YOLO is reliable alone

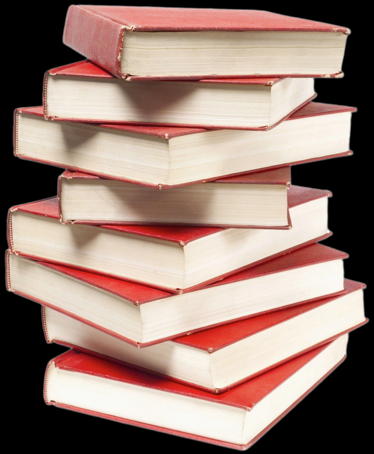
Transfer learning

- Enables strong results from only 485 training images

Template matching baseline

- Classical pixel-differencing approach failed (F1=0.004)
- Validates the need for deep learning

Literature Review



Object Detection
YOLO (Redmon 2016) introduced single-pass detection; YOLOv8 (Jocher 2023) adds to this research → we use YOLOv8 for real-time defect localization

Transfer Learning
ResNet (He 2015) pretrained on ImageNet (Deng 2009) → we fine-tune for 6-class PCB defect classification with limited training data

Classical CV
ORB features (Rublee 2011) + RANSAC alignment (Fischler & Bolles 1981) → our template matching baseline for comparison

Quality Management
1-10-100 Rule (Labovitz & Chang 1993), validated by NASA/INCOSE (Stecklein 2004) → frames our business case for early defect detection

PCB Datasets:
Kaggle PCB Defects (Akhatova), DeepPCB (Tang et al.) → 1,386 + 1,500 images; 6 defect types

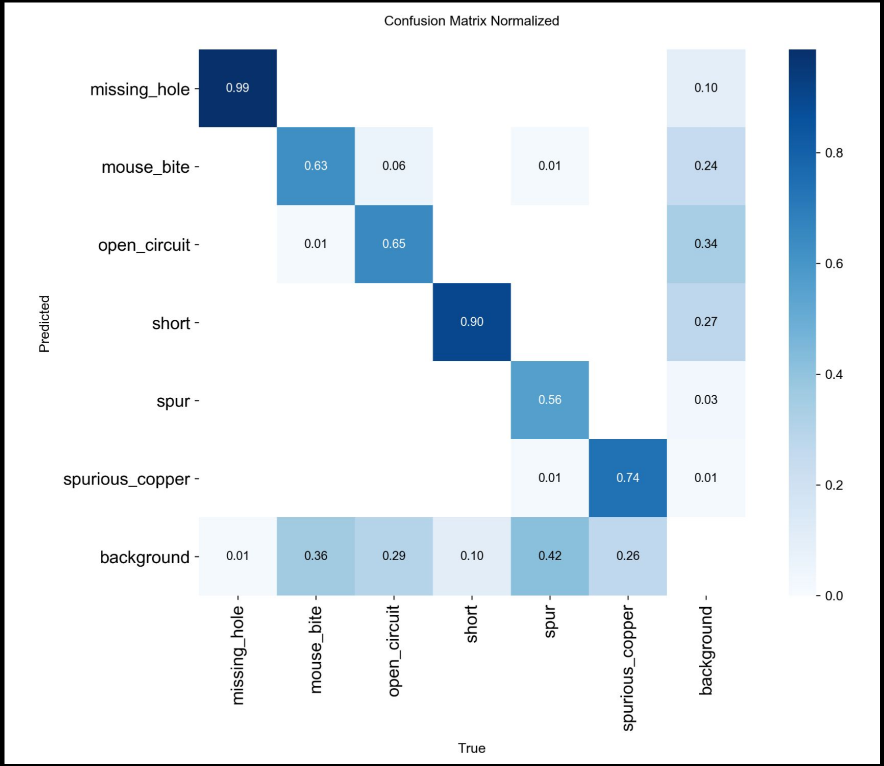
Results

Approach	Task	Precision	Recall	F1	mAP@0.5
YOLOv8	Detection + Classification	0.82	0.75	0.78	0.80
ResNet	Classification only (ground truth crops)	1.00	1.00	1.00	N/A
Template Matching	Detection only	0.002	0.01	0.004	N/A

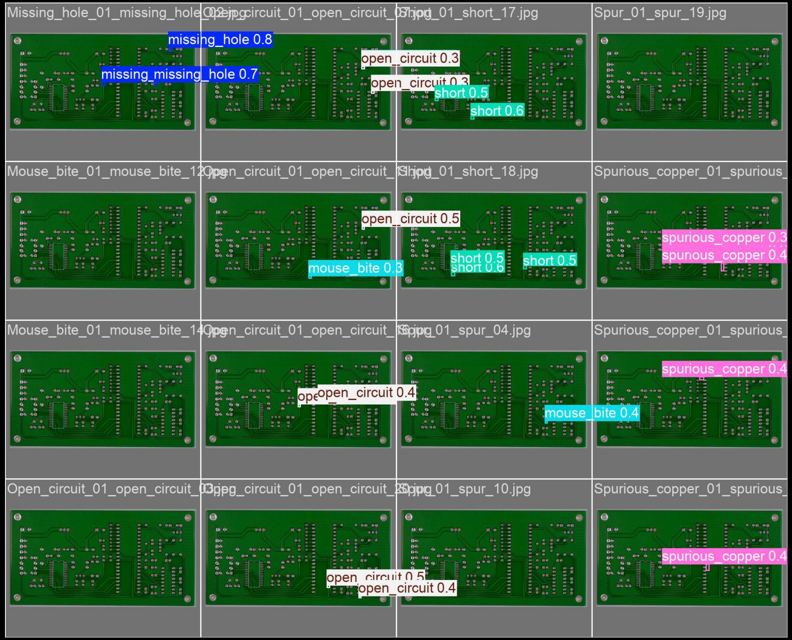
Two-stage pipeline (YOLO → ResNet) tested: 98.1% class agreement across 426 detections. Added ~44ms latency with negligible gain. Single-pass YOLO recommended.

YOLOv8 is the only approach that solves the full detection + classification task. Template matching validates the need for deep learning.

Results



Confusion Matrix Normalized
missing_hole
mouse_bite
open_circuit
short
spur
spurious_copper
background
Predicted
True
0.99
0.10
0.63
0.06
0.01
0.24
0.01
0.65
0.34
0.90
0.27
0.56
0.03
0.01
0.74
0.01
0.01
0.36
0.29
0.10
0.42
0.26
0.0
0.2
0.4
0.6
0.8



Spur_01_spur_19.jpg
Shor_01_short_18.jpg
Spur_01_spur_04.jpg
Spur_01_spur_10.jpg
missing_hole 0.8
missing_missing_hole 0.7
open_circuit 0.3
short 0.5
short 0.6
open_circuit 0.5
mouse_bite 0.3
spurious_copper 0.4
mouse_bite 0.4
open_circuit 0.4

Next Steps & Path to MVP

1

DONE

- Model inference
- Multi-technique evolution
- ONNX optimization
- Flask inference API

2

NEXT

- Docker containerization
- CI/CD pipeline
- Real-time camera integration
- Per-board fine-tuning on custom data
- Improve recall on weak classes

3

FUTURE

- Web dashboard for batch inspection
- Active learning (human review)
- Edge deployment
- Multi-board dataset

Thank you